BiRect: Bidirectional Rectangular Coverage Path Planning for a Swing-Arm Concrete Troweling Robot

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Abstract

Compared with center-mounted and front-mounted configurations, a swing-arm robot can keep the base clear of walls, columns, and openings while the disc covers lateral regions and narrow passages during large-area, repetitive troweling. Existing coverage path planning methods still focus on regular sites or agricultural fields with a fixed tool footprint; they do not account for an active swing arm or for the time-varying disc sweep on a complex map. This paper therefore presents BiRect. It proposes, for a known two-dimensional obstacle map, joining the visitation order over a complex free space and coordinated base and swing-arm motion into one executable route. It introduces bidirectional rectangular candidates, together with hole filling and connectivity completion, so that an irregular free space becomes rectangles that a swing arm can visit continuously. It develops local sweeping that varies with arm angle; the coverage ratio and execution time are computed from the actual sweep of the circular disc along the complete trajectory, while the coupled effects of base speed and arm angular velocity on sweep continuity and execution time are further revealed. Experiments showed that the proposed method achieves full-trajectory coverage ratios of 94.11% and 95.73%, respectively, and both generated routes pass the motion-feasibility checks. By contrast, the connecting segments produced by a fixed-width baseline leave the valid workspace by 4.51 m^2 and 20.00 m^2, making those routes non-executable. Experiments further show that the bidirectional candidate strategy attains the highest mean full-trajectory coverage ratio (94.32%) and the lowest mean total execution time (2316.60 s). These results support integrating complex-map decomposition, active arm motion, and continuous swept-area evaluation within a single planning pipeline. Validation in dynamic construction environments and on a physical robot remains future work.

Keywords: coverage path planning; construction robotics; concrete troweling; swing-arm robot; area decomposition

1 Introduction

Construction work entails substantial safety and health risks. The U.S. Bureau of Labor Statistics reported 1,032 fatal occupational injuries in construction and extraction occupations in 2024; common events included falls, slips, trips, and transportation incidents involving construction vehicles or machinery [1]. Beyond traumatic injuries, work with silica-containing materials such as concrete and mortar can expose workers to high concentrations of respirable crystalline silica over long periods, increasing the risks of silicosis, lung cancer, and chronic obstructive pulmonary disease [2]. Concrete floor leveling and troweling further combine a large work area, repeated passes, and a narrow time window governed by concrete setting. Manual work therefore requires sustained physical exertion and repetitive motion. Field physiological measurements of concrete slab placing and finishing crews showed that most manual workers experienced high energy expenditure and cardiopulmonary demand, with a greater risk of physical fatigue than equipment operators [3]. Surface finish also depends on operator experience, proficiency, and moment-to-moment condition [4, 5]. Releasing workers from hazardous, repetitive, and physically demanding floor operations while improving the consistency of the finished surface is therefore a problem of direct engineering and social value for construction robotics.

Floor-troweling robots provide a practical route toward mechanization. Early field trials showed that a remotely operated troweling robot could perform multiple operations from wood floating to steel troweling.

It treated approximately 1,000 m^2 across five sites and achieved an average productivity of about 317 m^2/h under the reported test conditions [5]. For an order-of-magnitude comparison, the average production rate of manual steel troweling is approximately 13.9 m^2/h [6]; the reported robotic rate is therefore about 22.8 times the rate of one manual worker. Subsequent work introduced electric actuation, laser localization, task planning, and remote monitoring to ride-on power trowels, allowing them to follow specified paths within a predefined floor area [7].

Most existing troweling robots use a center-mounted or front-mounted disc. Because the tool remains fixed relative to the base, the base path largely determines the tool path, and back-and-forth routes can be planned with a fixed working width [5, 7]. Swing-arm configurations have recently attracted attention because they enlarge the end-disc workspace without requiring a large base. Such a robot can place its circular disc beside the base or into locally constrained regions while the base remains clear of walls, columns, and openings, thereby partially decoupling the traversable base path from the tool trajectory. The arm can also increase the effective coverage width. For a fixed-disc platform, the nominal width of a straight pass is mainly limited by the disc diameter. A swing-arm platform can instead move the disc laterally while the base advances, so that the effective width depends on the arm length, angular range, and disc diameter and can exceed the disc diameter [8]. At the same base speed, provided that the swing frequency keeps consecutive sweeps contiguous without introducing gaps, the theoretical covered area per unit time can increase from approximately $v(2R)$ for a fixed disc to vw_{eff} for a swing arm, where $w_{\text{eff}} > 2R$. Previous coverage-planning research likewise shows that repeated coverage, average velocity, and the smoothness of turns jointly affect task time and energy consumption [9, 10]. The swing-arm configuration may also reduce unnecessary driving and sharp turns on wet concrete and provide more admissible poses near corners, narrow-passage entrances, and obstacles [8].



Figure 1: A swing-arm troweling robot operating on site. The tracked base, swing arm and actuator, and circular end disc are visible. Image source: Shangguan News; photograph provided by China Construction Eighth Engineering Division.

Mechanization and teleoperation, however, do not by themselves provide autonomy. A human teleoperator must continuously observe the floor and machine state while coordinating translation, steering, and troweling. In spaces divided by walls, columns, embedded components, and openings, occluded views and fragmented passages increase the burden of global route selection and can cause backtracking, detours, missed regions, and repeated work. Experiments with teleoperated mobile robots also show that manual driving time increases with environmental difficulty and depends on operator skill and the

level of obstacle-avoidance assistance [11]. Existing swing-arm systems commonly rely on human–robot cooperation or predefined trajectories and do not yet generate complete coverage routes automatically for complex construction maps [8]. Continuous teleoperation also requires a skilled operator to remain on duty, adding training and on-site supervision costs. Empirical studies of construction-robot adoption further identify economic factors, work culture, and uncertain business returns as barriers to deployment [12, 13]. Without autonomous coverage planning, these operational burdens remain even after the troweling process is mechanized.

This task is an instance of coverage path planning (CPP): the tool must sweep the target area while avoiding obstacles and controlling missed coverage, repeated coverage, transfer distance, and execution time [9]. Existing approaches include back-and-forth sweeps after cellular decomposition [14, 15], decomposition and transfers on polygons with holes [16], grid or spanning-tree traversal [17, 18], and strip generation with curvature-constrained connections for fixed-width implements in applications such as agriculture [19, 20]. Mobile-manipulator methods can plan local motions for a base and an end tool [21], while SatAOI can identify the workable area of a swing-arm troweling robot in an obstacle map [8]; neither, however, provides a complete continuous coverage route for a complex two-dimensional floor. Section 2 gives a detailed classification and discusses these boundaries.

CPP for a swing-arm troweling robot introduces three interdependent requirements. First, the planner must represent a complex free space containing openings, dense obstacles, and narrow connections. Second, at a fixed base pose, it must distinguish the disc positions and whole-robot collision states associated with different arm angles. Third, while the base moves continuously, actual coverage must be determined from arm angular velocity and the continuous disc sweep rather than from a static reachability envelope. A static reachability envelope is the region the arm can touch in some poses and does not include the time course of base travel and swing-arm motion. Simply dilating the base path into a constant-width strip cannot satisfy all three requirements. Consequently, prior work lacks a complete planning chain from a construction map through decomposition and visitation order to coordinated base–arm motion and end-tool coverage evaluation.

This paper proposes BiRect, a hierarchical coverage path planning method that places bidirectional rectangular decomposition, the visitation order on a feasible transition graph, and the coverage ratio and execution time computed from the area actually swept by the circular disc in one chain, using the same base–swing-arm–disc model for the map, local motion, and trajectory evaluation.

The remainder of the paper is organized as follows. Section 2 reviews the development of concrete-troweling robots and the challenges posed by obstacle-rich construction environments, compares representative CPP methods, and summarizes the unresolved research gap. Section 3 presents the robot and environment models, rectangular-candidate decomposition, connectivity completion, global visitation, and swing-assisted coverage within each rectangle. Section 4 evaluates the method through a baseline comparison, candidate-scan strategy experiments, and a speed-sensitivity study. Section 5 summarizes the experimentally supported contributions, limitations, and future work.

2 Related Work

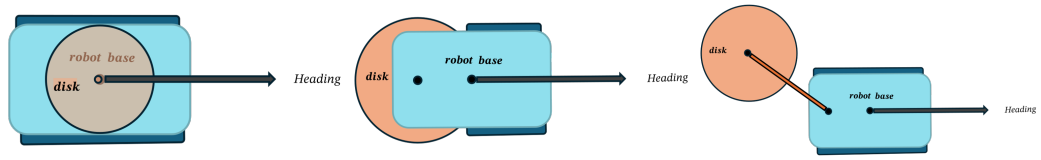
2.1 Concrete Troweling Robots and Application Challenges

After concrete has been placed and initially leveled, it must be troweled repeatedly before setting and hardening. Troweling removes local height variations and rough texture, redistributes the surface paste, and produces a flat, dense finish through compaction and polishing. The operation covers a large area and is highly repetitive, while its start time, speed, and number of passes are constrained by the state of concrete setting. Starting too early can disturb an unstable surface; starting too late can prevent the desired flatness and finish. Site practice has therefore long depended on the experience and continuous involvement of skilled workers [4, 5]. A troweling robot moves the disc and treats the surface at a relatively stable speed and cadence, reducing physical workload and the effect of worker-state variability on quality

consistency [5, 22]. Its role is thus not merely to replace a hand tool, but to convert a process that requires continuous manual operation into one that a mechanical system can execute consistently.

The evolution of concrete-troweling equipment can be understood along two axes: the disc–base relationship and the mode of control.

Along the disc–base relationship, existing troweling robots fall into three configurations. A center-mounted disc lies under or near the center of the base, as on the ride-on twin-rotor machine [7] and on the twin-disc machine whose discs sit under the body and whose travel follows disc attitude [22]. A front-mounted arrangement places troweling discs at both the front and the rear of the travel mechanism; the discs stay fixed relative to the base and have no active lateral motion [5]. A swing-arm configuration places the disc at the end of an actively rotating arm [8]. The swing arm can send the disc into regions beside walls, columns, openings, and narrow passages while the base stays clear of them. When consecutive sweeps remain contiguous, the same base speed can give an effective coverage width larger than the disc diameter. This configuration has received increasing attention in recent construction-robot research [8]. These advantages come from coordinating base translation with swing-arm motion and cannot be represented by a fixed-width strip model.



(a) Underbody or center-mounted disk (b) Front-mounted disk (c) Swing-arm-mounted disk

Figure 2: Three arrangements of the troweling disc relative to the base: (a) center-mounted; (b) front-mounted; and (c) swing-arm. Arrows indicate the direction of base travel.

The state variables x_c and y_c locate the center of the base in the ground frame, θ is the base heading, and ϕ is the arm angle relative to the base. Together, the four variables determine the poses of the base, swing arm, and disc in the construction plane. A region traversable by the base is not necessarily reachable by the disc, and a disc-reachable pose is not necessarily collision-free for the complete mechanism. Planning must therefore check the geometry of all three components and coordinate the base and swing arm throughout continuous motion [8].

Along the mode of control, the equipment falls into three modes: manual walk-with or ride-on, wireless teleoperation, and autonomous travel in regular areas. Walk-behind and ride-on power trowels first mechanized the operation, but a worker still walked with or rode on the machine. Because the disc remained fixed relative to the base, the machine route effectively determined the disc trajectory [7]. Wireless teleoperation then moved the operator outside the work area. Kobayashi et al. fixed front and rear troweling discs to a lightweight mobile base and used wireless teleoperation and taught-motion replay to perform successive operations from wood floating to steel troweling [5]. The system reduced worker exertion, but the discs still moved rigidly with the base, and coverage remained sensitive to operator experience, floor friction, and steering behavior [5]. Autonomous travel in regular areas came next. Hwangbo et al. retained the fixed twin-rotor troweling mechanism of a ride-on machine, electrified controls such as steering, blade pitch, and acceleration, and added laser localization, reflective landmarks, and a construction-drawing-based task planner. The resulting machine could work autonomously along a prescribed pattern or planned route [7]. This improved control autonomy without changing the rigid coupling between disc and body.

A swing arm makes the disc movable relative to the base, so autonomous travel with a fixed disc is

not enough for the platform studied by SatAOI. SatAOI starts from an obstacle grid, models the base, swing arm, and disc, and searches for connected regions that the robot can enter and in which the disc can operate [8]. Its output stops at that workable area: it gives no visitation order, no base transfer between regions, and no coordinated base–swing-arm coverage inside a region. This paper starts from that area and plans a continuous coverage route.

2.2 Coverage Path Planning Methods

Existing CPP methods can be compared along two dimensions: map-representation capability and fidelity of robot-behavior modeling. The first describes whether a planner can represent irregular boundaries, multiple obstacles, holes, and narrow connections and use them to establish adjacency and feasible transfers between coverage regions. The second describes whether the planner captures the physical dimensions, relative poses, motion modes, and collision constraints of the robot and tool rather than replacing the operation with a point, disc, or fixed-width strip. These dimensions define four capability classes, and the names below are those used in this paper. Idealized coverage methods are weak in both map representation and robot-behavior fidelity, and they mainly establish fundamental coverage mechanisms. Map-enhanced methods represent a complex map but keep a fixed tool footprint. Platform-constrained methods describe motion constraints in detail but primarily target regular or local workspaces. Environment–platform coupled methods retain both a complex map and realistic robot behavior.

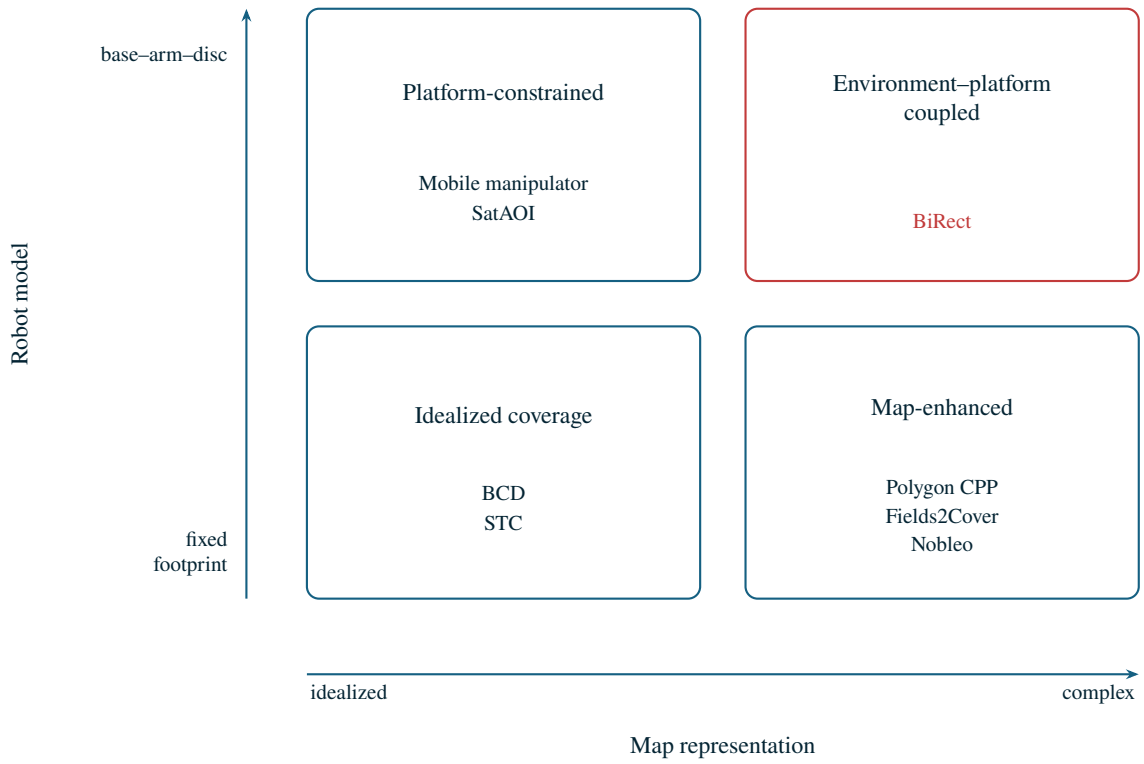


Figure 3: Four capability classes arranged by map representation and robot model.

The first class comprises classical coverage algorithms that express the core coverage idea clearly but generally close the problem only for idealized maps and simplified robot models. Representative methods include Boustrophedon Cellular Decomposition (BCD) and Spanning Tree Coverage (STC). BCD scans the free space and ends the current cell when the number of free intervals on the scan line changes, then generates back-and-forth sweeps inside the cell [14]. STC coarsens the free space into a grid, builds a spanning tree on that grid, and walks the fine cells around the tree boundary [17]. These methods establish

the respective roles of cellular decomposition and graph traversal. They do not represent the relative position of the base and the disc, nor the change of the arm angle ϕ, and are therefore difficult to apply to swing-arm troweling, on which the disc moves relative to the base.

The second class can represent complex maps, but the robot and tool are still modeled mainly by a fixed footprint. It first represents a map with holes, non-convex boundaries, or obstacles, then covers that map with fixed-width strips, orders the regions, and connects them by collision-free turns. Reproducible examples include ETH Zurich's `polygon_coverage_planning` [15, 16], Fields2Cover for agricultural vehicles [19, 20], and the Nobleo Full Coverage Path Planner for ROS occupancy grids [18]. `polygon_coverage_planning` handles general polygons with holes, uses BCD or trapezoidal decomposition to generate cells, constructs multiple sweep candidates for each cell, and performs global selection and collision-free transfer through an Equality Generalized Traveling Salesman Problem and a visibility graph. Fields2Cover supports non-convex fields, obstacles, headlands, swaths, and route ordering, and connects swaths by curvature-constrained Dubins or Reeds–Shepp connections [19, 20]. Nobleo's planner implements a Backtracking Spiral Algorithm/Spiral-STC-type method [18] on a ROS occupancy grid and permits separate robot and tool radii. These frameworks show that complex boundaries, obstacles, and engineering interfaces can be addressed systematically. Their tool width, implement width, or tool disc nevertheless remains essentially fixed relative to the base and cannot represent the time-varying sweep caused by continuous changes in ϕ.

The third class retains, during planning, the minimum turning radius, the reachable range of the arm, and component collisions, while its scenes are mostly regular sites, local surfaces, or three-dimensional objects. It includes the nonholonomic vehicle connections in Fields2Cover and mobile-manipulator surface-coverage methods for tasks such as spray disinfection. Fields2Cover incorporates minimum turning radius and vehicle kinematics [19, 20], but targets agricultural swaths with a fixed implement width. Mobile-manipulator studies explicitly handle the base, arm, end tool, reachability, and occlusion. For example, spray-based surface disinfection can generate end-effector paths from three-dimensional surfaces and jointly select base positions and arm motions [21]. Such methods describe how a robot executes a task in greater detail, but generally target regular sites, agricultural strips, or three-dimensional object surfaces rather than jointly performing obstacle handling, area decomposition, global traversal, and continuous ground-coverage evaluation on a known two-dimensional construction map.

Swing-arm troweling therefore requires the fourth class shown in Fig. 3. Walls, columns, reserved openings, and narrow corridors divide the floor into irregular regions with complex connectivity, which the planner must represent and visit continuously. Meanwhile, the disc position is jointly determined by base motion and active swing-arm motion; coverage and collision status vary continuously with base heading and arm angle and cannot be approximated by a fixed footprint. A complete, continuous, and executable route therefore requires the construction map and the coordinated base–swing-arm–disc behavior to be represented in the same planning model.

Research has thus advanced in both complex-map representation and robot-behavior modeling, but the two have not yet been unified, especially for swing-arm floor coverage. CPP frameworks with strong map support usually assume a fixed tool footprint and cannot describe the dynamic coverage created by a rotating arm. Methods that model vehicle kinematics, manipulator reachability, and component collisions in detail generally focus on regular sites, local operations, or three-dimensional surfaces and do not fully decompose and continuously traverse a complex two-dimensional construction map. SatAOI considers both an obstacle map and the geometry of the base, arm, and disc, but stops at delimiting the workable area and does not produce a visitation order, transfer route, or coordinated coverage trajectory within each region [8]. The central gap is therefore a swing-arm troweling CPP method that combines complex-map processing with high-fidelity robot-behavior modeling and generates a complete, continuous, executable route directly from the construction map.

Table 1 compares representative approaches along these two dimensions. Work spanning more than one class is placed according to its principal contribution: Fields2Cover is treated as map-enhanced and SatAOI as platform-constrained. The stated limitations concern direct application to a mobile base with an active arm and circular disc; they are not general judgments of the methods or software.

Table 1: Representative CPP methods and their applicability to swing-arm troweling.

Category	Representative method	Map representation	Robot model	Pros and cons
Idealized coverage	BCD [14]	Cellular decomposition	Fixed footprint	✓ Organizes back-and-forth sweeps within cells. × Does not represent an active arm or continuous sweep.
Idealized coverage	STC [17]	Grid spanning tree	Point or disc robot	✓ Systematically traverses a grid. × Cannot use the time-varying disc position created by arm motion.
Map-enhanced	Polygon CPP [15, 16]	Polygon with holes	Fixed footprint	✓ Provides alternative strips and obstacle-avoiding transfers. × Does not represent the arm's instantaneous disc position.
Map-enhanced	Fields2Cover [19, 20]	Non-convex field	Fixed footprint	✓ Includes swaths, route ordering, and curvature-constrained connections. × The fixed footprint omits arm motion.
Map-enhanced	Nobleo FCPP [18]	Occupancy grid	Fixed footprint	✓ Integrates grid coverage with a navigation stack. × Does not coordinate arm angle, arm angular velocity, and base speed.
Platform-constrained	Mobile-manipulator surface coverage [21]	Three-dimensional surface	Mobile manipulator	✓ Jointly plans base placement and end-tool motion. × Does not completely traverse a complex two-dimensional floor.
Platform-constrained	SatAOI [8]	Construction obstacle map	Base, arm, and disc	✓ Determines workable regions and connectivity. × Does not generate a visitation order or continuous coverage trajectory.
Proposed method	BiRect	Two-dimensional obstacle map	Swing arm with a time-varying disc	✓ Links rectangular decomposition, visitation, local swinging, and trajectory evaluation. × Currently assumes a static map and uses heuristic search.

2.3 Summary and Research Gap

Prior work provides several capabilities needed here, but has not solved complex-map coverage within one unified task. BCD [14] and STC [17] can represent two-dimensional free spaces and can decompose regions or traverse grids. Decomposition and swath generation on regions with holes or non-convex boundaries are given in [15, 16], and grid traversal in [18]. With a tool fixed to the base, they can

arrange adjacent sweeps according to a prescribed width. Fields2Cover additionally generates turns and curvature-constrained connections for vehicles whose direction and minimum turning radius prevent arbitrary lateral motion [19, 20]. Mobile-manipulator research addresses end-tool reachability and collision relationships [21], while SatAOI uses the geometry among the base, swing arm, disc, and obstacles to delimit regions the robot can enter and the disc can treat [8]. These abilities remain distributed across different task models and applications: complex-map CPP usually assumes a fixed footprint, detailed robot models are commonly used for local or regular workspaces, and SatAOI terminates at workable-area delimitation. For the specific objective of generating a complete executable route for a swing-arm troweling robot in a known, complex, two-dimensional construction map, three coupled gaps remain.

First, existing CPP methods do not model active swing-arm behavior sufficiently to plan the actual coverage of the disc: in conventional CPP the tool is fixed to the base, so determining the base route also determines coverage. At the same base position and heading, a different arm angle places the disc center on an arc whose radius is the arm length, and the whole-robot collision state differs as well. Maximum arm reach, or a widened strip, only shows that some pose can reach a location; it is not coverage during continuous motion.

Second, complex-map decomposition and local swing motion have not been integrated into a computationally practical planning chain. SatAOI stops at the workable area; this paper uses a hierarchical heuristic to join region connectivity, visitation order, and coordinated base–swing-arm coverage inside each region into one computable chain.

Third, a static reachability envelope does not guarantee complete coverage during dynamic execution, and prior work does not explicitly model base–arm velocity coupling, the coordination of base linear velocity and arm angular velocity. Maximum arm reach indicates only which locations the disc can touch in some poses. While the base advances continuously and the swing arm swings periodically, the longitudinal spacing between consecutive sweeps is determined by base speed and arm angular velocity; if the base is too fast or the swing arm too slow, untreated gaps remain even when the target lies inside the static reachability envelope defined above, so being reachable is not the same as being covered.

3 Methodology

This study addresses the planning problem that maps a known, static, two-dimensional obstacle environment to a complete coverage trajectory for a swing-arm troweling robot. Walls, columns, and openings divide a construction floor into a free space with complex geometry and connectivity, while the disc position and footprint are jointly determined by the base pose and arm angle. The planner must therefore organize global visitation of the free space and convert coverage of each region into continuous, coordinated base–arm motion. Subject to obstacle-avoidance, geometric, and motion constraints, the objectives are to increase effective coverage of the target floor and reduce both trajectory execution time and planning time. The number of rectangles, their overlap, and the number of inter-rectangle transfers are intermediate measures only; they are converted into proxy costs for polynomial-time heuristics at the respective planning levels.

3.1 Overall Planning Pipeline

BiRect proceeds in four stages, and the output of each stage is the input of the next. Figure 4 summarizes the workflow. To improve the efficiency of searching a complex free space, rectangular decomposition gathers that space into a small number of bidirectional rectangular candidates. Connectivity completion outputs a feasible adjacency graph G, whose edges mean that a base–swing-arm transfer satisfying the geometric and velocity constraints exists, not merely that two rectangles touch. Then each rectangle supplies the time of a first coverage visit and of a repeated transit visit, and these times are used as the costs of global route planning. The global stage obtains a visitation sequence on G and joins it into a trajectory; evaluation reads only this trajectory and returns the coverage ratios and the execution time, while T_{plan} is the computation time of the whole workflow and is not used as a search cost.

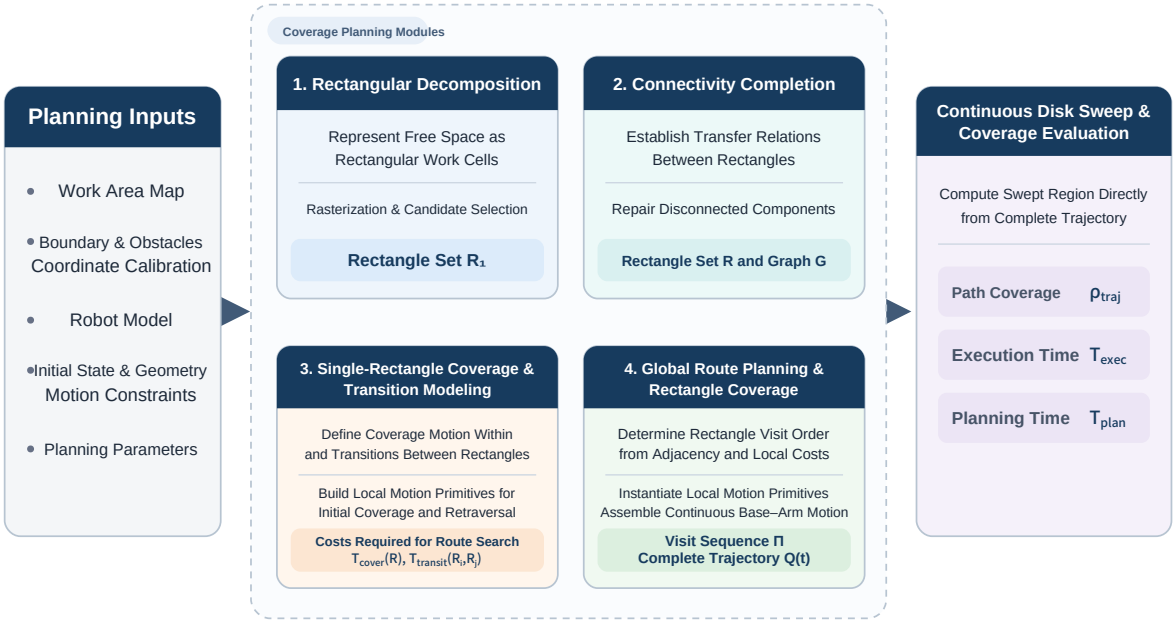


Figure 4: Workflow of BiRect.

3.2 Problem Formulation and Notation

3.2.1 From a Construction Scene to Planning Inputs

The required scene information is first extracted from a CAD/BIM model, site-survey data, or localization and mapping output. For a floor to be troweled, a world-coordinate plane $\mathcal{F}_W$ coincident with the work surface is selected, and the boundary of the current task is specified in that plane. Walls, columns, equipment foundations, embedded components, reserved openings, and areas outside the work boundary are marked as unavailable. The remaining floor inside the boundary on which the disc may operate becomes the target work region. The classified plane is then rasterized and geometrically calibrated to produce a physically scaled binary work map M, which is the direct map input to CPP.

At a resolution of s_p m/pixel, blue pixels in result and evaluation images denote portions of the target region already swept by the disc. Blue is an overlay on M, not a third input-map class. A rigid transform T_{WM} calibrates image coordinates to world coordinates, so every pixel corresponds to a physical location in metres.

Unavailable regions are first enlarged by the safety margin δ_{map} and then rasterized, and every boundary cell or cell of ambiguous membership is treated as unavailable. Circular columns, curved boundaries, and openings become staircase boundaries at the selected resolution. Straight walls and rectangular foundations map to contiguous occupied cells.

After the map has been abstracted, equipment drawings, calibration tests, and controller limits provide the robot model and initial state. The scene input for one planning task is divided into map and robot components:

$$\mathcal{I} = \left\langle \mathcal{I}_{\text{map}}, \mathcal{I}_{\text{robot}} \right\rangle,$$

where

$$\mathcal{I}_{\text{map}} = \left\langle M, s_p, T_{WM} \right\rangle.$$

Here M is the $H_p \times W_p$ binary work map generated from CAD/BIM, survey, or mapping data; s_p converts pixel distance and area to metres and square metres; and T_{WM} is the planar rigid transform from the map frame to the world frame, used to convert planned map coordinates into executable site coordinates.

The robot component is

$$\mathcal{I}_{\text{robot}} = \langle q_0, \Theta_R \rangle.$$

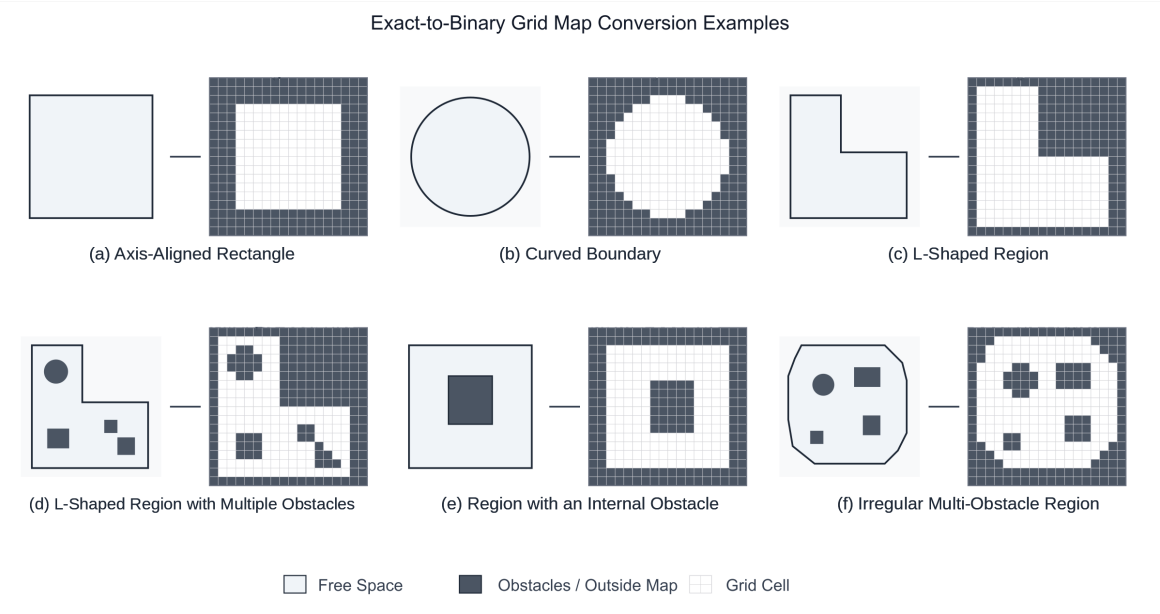


Figure 5: Conversion from exact geometric descriptions to the binary raster input.

The initial base position, heading, and arm angle in q_0 are obtained from the site-localization system and arm encoder. The parameter set Θ_R contains the base length l_c and width w_c, arm-link collision radius r_a, troweling-disc radius r_d (diameter $2r_d$), arm length L, distance d_p from the base center to the arm pivot, arm-angle limits, maximum base linear and angular velocities, and maximum arm angular velocity. Together, these parameters determine occupied geometry, the disc coverage footprint, and trajectory executability.

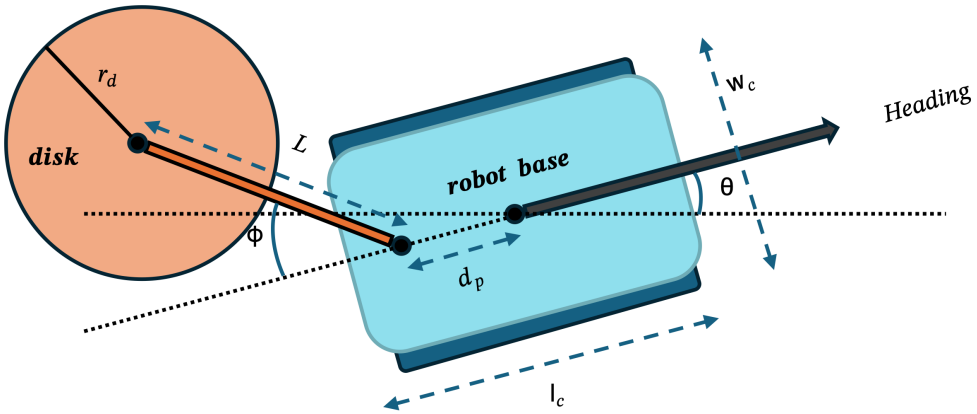


Figure 6: Geometry and motion parameters of the base–arm–disc system.

In addition to these scene inputs, the algorithm uses a planning configuration Ψ_P: planning-cell size g, minimum rectangle side length, candidate-selection parameters α and β, candidate truncation count, scan and connectivity strategies, and random seed. The configuration controls decomposition and search but is not part of the map or robot model abstracted from the physical scene.

3.2.2 Planning Model and Notation

Let the input map be

$$M \in \{0, 255\}^{H_p \times W_p},$$

The labels 0 and 255 only distinguish obstacle and free; they are not grayscale values and not physical quantities. With physical pixel size s_p, the free-pixel set is

$$\mathcal{P}_F = \{(u, v) \mid M_{uv} = 255\}.$$

All subsequent positions and distances are transformed by T_{WM} into a metric world frame; for clarity, the transform is omitted from the remaining notation.

The robot state is

$$q(t) = \big(x_c(t), y_c(t), \theta(t), \phi(t)\big),$$

with state space $\mathcal{Q} = \mathbb{R}^2 \times S^1 \times [\phi_{\min}, \phi_{\max}]$. Here (x_c, y_c) is the base center, θ the base heading, and ϕ the arm angle relative to the base. Let $\mathbf{p}_c = (x_c, y_c)$ and $\mathbf{e}(\gamma) = (\cos\gamma, \sin\gamma)$. For arm length L and base-center-to-pivot offset d_p, the center of the circular disc is

$$\mathbf{p}_d(q) = \mathbf{p}_c - d_p\mathbf{e}(\theta) + L\mathbf{e}(\theta + \pi + \phi). \tag{1}$$

Equation (1) is the sum of two translations: from the base center to the arm pivot and from the pivot to the disc center. As shown in Fig. 6, $\mathbf{e}(\theta)$ points along the base heading, and the pivot lies d_p behind the base center, so

$$\mathbf{p}_p = \mathbf{p}_c - d_p\mathbf{e}(\theta).$$

The arm angle ϕ is defined as the signed rotation from the rearward base axis, positive counterclockwise. Since the rearward axis is rotated by π from the forward heading, the pivot-to-disc direction is $\theta + \pi + \phi$, and its displacement is $L\mathbf{e}(\theta + \pi + \phi)$. For example, when $\theta = 0$ and $\phi = 0$, the base faces the positive world x-axis, the pivot is at $(x_c - d_p, y_c)$, and the disc center is at $(x_c - d_p - L, y_c)$: the arm extends directly behind the base. Changing θ rotates the pivot and arm with the base; changing ϕ at fixed θ moves the disc center along a radius-L arc about the pivot.

For disc radius r_d, the instantaneous work region at one pose is

$$\mathcal{D}(q) = \{\mathbf{z} \in \mathbb{R}^2 \mid \|\mathbf{z} - \mathbf{p}_d(q)\|_2 \le r_d\}. \tag{2}$$

The region swept by a continuous trajectory $q : [0, T] \to \mathcal{Q}$ is

$$C(q) = \bigcup_{t \in [0,T]} \mathcal{D}(q(t)). \tag{3}$$

To distinguish disc reachability from whole-robot feasibility, let $\mathcal{O}_M$ be the unavailable region corresponding to black pixels in world coordinates, $\mathcal{B}(q)$ the rotated $l_c \times w_c$ base rectangle, and $\mathcal{A}(q)$ a capsule of radius r_a around the arm link. A state is geometrically feasible only if

$$\big(\mathcal{B}(q) \cup \mathcal{A}(q) \cup \mathcal{D}(q)\big) \cap \mathcal{O}_M = \varnothing. \tag{3a}$$

All feasible states form $\mathcal{Q}_F \subseteq \mathcal{Q}$. For a motion segment $q_s : [0, T_s] \to \mathcal{Q}$, the method requires $q_s(t) \in \mathcal{Q}_F$ for every $t \in [0, T_s]$ and

$$|v(t)| \le v_{\max}, \qquad |\dot\theta(t)| \le \omega_{\max}, \qquad |\dot\phi(t)| \le \omega_{a,\max}. \tag{3b}$$

Straight motion, in-place rotation, arm adjustment, and other primitives are adaptively subdivided, and Eq. (3a) is checked over every subinterval. Consequently, later rectangle adjacencies and entry states are candidates only; a transfer enters the final planning graph only after passing this check.

Under Eqs. (3a) and (3b), the algorithm is assessed at three levels: effective-work coverage, trajectory-execution efficiency, and planning efficiency. The corresponding objectives are

$$\max \rho_{\text{work}}(q), \qquad \min T_{\text{exec}}(q), \qquad \min T_{\text{plan}}. \tag{4}$$

The effective-work coverage ratio $\rho_{\text{work}}(q)$ is the fraction of the target area swept by the disc during work-labeled segments. The nominal execution time $T_{\text{exec}}(q)$ runs from the initial state to completion and includes the initial connection, effective work, turns, arm adjustments, and inter-region transfers. The wall-clock planning time T_{plan} covers the algorithm from conservative-grid generation through trajectory assembly and coverage evaluation.

These measures play distinct roles. The task-level coverage measure ρ_{work} is encouraged indirectly through free-space rectangle representation, rewards for new coverage, and overlap penalties, then computed from the continuous disc sweep in Eqs. (1)–(3) after the complete trajectory is available. The search uses time proxies of local coverage, transit, and transfer to compare visitation sequences; the full execution time T_{exec} is computed after the trajectory has been generated and is not used as a cost in the search. The algorithm selects feasible trajectories with low proxy cost but does not claim a global minimum. The planning-time measure T_{plan} is not a decision variable of q; it is a design objective controlled by hierarchical decomposition, polynomial-time heuristics, and candidate truncation. This separation motivates the hierarchical heuristic framework.

3.3 Conservative Rasterization and Rectangular Decomposition

Following the conservative rule of Section 3.2.1, the test is written here on planning cells. Let each square planning cell contain $g \times g$ pixels, and let Ω_{ij} be the pixel block associated with planning cell (i, j):

$$B_{ij} = \begin{cases} 1, & \text{if } M_{uv} = 255, \quad \forall (u, v) \in \Omega_{ij}, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

On B, the algorithm scans maximal contiguous runs of free cells by rows and by columns, extending them across adjacent scan lines as long as every enclosed cell remains free. This produces axis-aligned candidates of the form

$$R = [x_1, x_2] \times [y_1, y_2], \tag{6}$$

All planning cells inside R are free. Candidates whose width or height is below a specified minimum are removed. The remainder are ordered first by decreasing area and, for equal areas, by increasing row and column coordinates of the upper-left and then lower-right corner. Only the first $C_{\max}$ candidates from each scan direction are retained. This deterministic truncation controls rapid growth in candidate count while making the candidate set repeatable for fixed inputs and parameters.

Candidate selection is formulated as a penalized maximum-coverage problem. Let $\mathcal{U}(R)$ be the set of planning cells covered by R and $\mathcal{A}$ the cells already covered by selected rectangles. The marginal utility of selecting R is

$$\Delta(R \mid \mathcal{A}) = |\mathcal{U}(R) \setminus \mathcal{A}| - \alpha - \beta|\mathcal{U}(R) \cap \mathcal{A}|. \tag{7}$$

The absolute values in Eq. (7) are counts of planning cells, so α has units of cells and β is dimensionless. The first term rewards new coverage; the other terms penalize rectangle count and overlap. The algorithm repeatedly selects the row-scan candidate with the largest positive marginal utility, using the deterministic ordering above to break ties. It then processes the column-scan pool relative to the current $\mathcal{A}$, so column-oriented rectangles mainly supplement vertical structures poorly represented by row scans. The two greedy stages produce the initial selection.

Candidate truncation or the penalties may leave a small number of free cells uncovered. A deterministic hole-filling stage therefore selects the first uncovered free cell in row-major order as a seed, enumerates maximal axis-aligned obstacle-free rectangles containing the seed, and selects the largest. Remaining ties are resolved by rectangle width and upper-left coordinates. The selected rectangle is added and $\mathcal{A}$ updated until every usable planning cell belongs to at least one rectangle. The resulting set, before connectivity repair, is $\mathcal{R}_1$. Minimum side length, $C_{\max}$, α, β, and scan direction are included in Ψ_P.

3.4 Connectivity Completion

Covering every free planning cell with rectangles does not ensure that the robot can move continuously between them. The method therefore constructs an undirected graph

$$G = (V, E), \tag{8}$$

over $\mathcal{R}_1$, with node $v_i \in V$ representing rectangle R_i. Shared boundaries, contact, and overlap generate geometric adjacency candidates only. For each candidate pair (R_i, R_j), a finite set of handoff poses is generated in the intersection or along the closest opposing boundaries. The algorithm then attempts to connect an exit state of the first rectangle to an entry state of the second. Edge (v_i, v_j) is added only if at least one transfer primitive satisfies Eqs. (3a) and (3b). The edge stores the minimum time proxy c_{ij} among all feasible handoffs. Graph G consequently represents executable robot transfers, not merely geometric contact between rectangles.

If G has multiple connected components, the algorithm searches the unselected candidates for bridging rectangles. A candidate R_b is a bridge if, after insertion, it forms feasible edges with rectangles in at least two previously distinct components. Preference is given to rectangles with little new area and low feasible-transfer cost:

$$R_b^\star = \arg\min_{R_b \in \mathcal{B}} \left(|\mathcal{U}(R_b) \setminus \mathcal{A}| + \lambda_b \sum c_{ib} \right), \tag{9}$$

where $\mathcal{B}$ is the current bridge-candidate set, the sum contains only edges used to connect components, and λ_b weights transfer time. Relevant edges and connected components are recomputed after every insertion, until G becomes connected. The final rectangle set $\mathcal{R}$ is $\mathcal{R}_1$ after the bridging rectangles have been added. If $\mathcal{B}$ is empty while G remains disconnected, no single continuous coverage trajectory is supported by the current candidate set and parameters. The method reports connectivity failure and does not proceed to global route planning.

Algorithm 1 Rectangle segmentation and connectivity completion

Require: Conservative free-cell grid B, planning parameters Ψ_P, robot model Θ_R
Ensure: Rectangle set $\mathcal{R}$ and feasible graph G, or connectivity failure

$F \leftarrow \{u : B(u) = 1\}$; $\mathcal{R} \leftarrow \varnothing$; $\mathcal{A} \leftarrow \varnothing$
$(C_r, C_c) \leftarrow \text{ScanAndTruncate}(B, \Psi_P)$
$C_{\text{all}} \leftarrow C_r \cup C_c$
for $C \in (C_r, C_c)$, in this order **do**
  while $\exists R \in C : \Delta(R \mid \mathcal{A}) > 0$ **do**
    $R^\star \leftarrow \arg\max_{R \in C} \Delta(R \mid \mathcal{A})$ ▷ Eq. (7); deterministic ties
    $\mathcal{R} \leftarrow \mathcal{R} \cup \{R^\star\}$; $\mathcal{A} \leftarrow \mathcal{A} \cup \mathcal{U}(R^\star)$
    $C \leftarrow C \setminus \{R^\star\}$
  end while
end for
while $\mathcal{A} \neq F$ **do**
  $u \leftarrow \text{FirstUncovered}(F \setminus \mathcal{A})$
  $R^\star \leftarrow \text{LargestMaximalRectangle}(B, u)$ ▷ deterministic ties
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{R^\star\}$; $\mathcal{A} \leftarrow \mathcal{A} \cup \mathcal{U}(R^\star)$
end while
$\mathcal{R}_1 \leftarrow \mathcal{R}$; $G \leftarrow \text{FeasibleGraph}(\mathcal{R}_1, \Theta_R)$
while G is disconnected **do**
  $\mathcal{B} \leftarrow \text{FeasibleBridges}(C_{\text{all}} \setminus \mathcal{R}, G, \Theta_R)$
  if $\mathcal{B} = \varnothing$ **then**
    return connectivity failure
  end if
  $R_b^\star \leftarrow \text{MinBridgeCost}(\mathcal{B})$ ▷ Eq. (9)
  $\mathcal{R} \leftarrow \mathcal{R} \cup \{R_b^\star\}$; $G \leftarrow \text{UpdateFeasibleGraph}(G, R_b^\star)$
end while
return $(\mathcal{R}, G)$

Algorithm 1 summarizes the complete process from conservative rasterization to the final graph. The overlap penalty in Eq. (7) suppresses redundant rectangles within a candidate pool; hole filling represents every free planning cell; and bridging adds rectangles only when the current set cannot form a connected feasible graph. "Connected" always refers to graph connectivity after complete robot-transfer checks, not merely topological connectivity of free cells or the geometric rectangle union.

Let the planning grid be $H \times W$, the truncated candidate count C, the final rectangle count n, and the mean candidate area A cells. Candidate generation has worst-case complexity $O(HW(H+W))$. With a naive set implementation, greedy selection is approximately $O(CnA)$. Adjacency-candidate enumeration is $O(n^2)$, with a finite number of local-motion checks per edge. The minimum side length, candidate truncation, and deterministic ordering jointly limit practical problem size. Since later graph search and local-primitive instantiation grow with n, controlling the rectangle count during decomposition directly affects T_{plan}.

3.5 Single-Rectangle Coverage Primitive

The rectangle-coverage module supplies the local motion primitive. Given a rectangle R, feasible entry pose q^{in}, feasible exit pose q^{out}, and robot parameters, it generates a continuous motion segment. On the first occurrence of R in the global sequence, the robot executes the back-and-forth coverage shown in Fig. 7; if R has already been covered, only an entry-to-exit transit motion is generated. Both primitive types retain the complete robot state, duration, and work-state label.

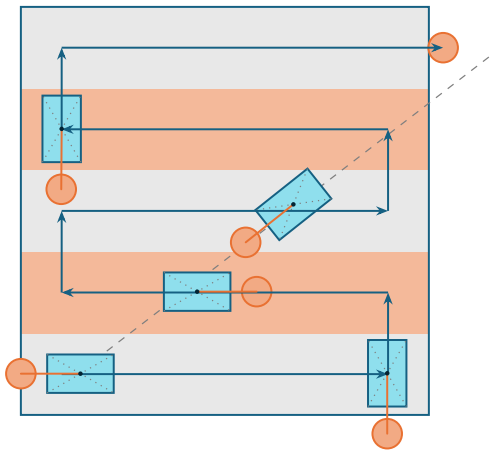


Figure 7: Single-rectangle coverage primitive.

Local paths are not redrawn for other aspect ratios. Each local path is a back-and-forth strip along one axis of the rectangle: the longer side is the default so as to reduce lane changes, and the other axis is also generated and compared by time.

3.5.1 Entry and Exit Poses

Entry and exit candidates are jointly determined by the predecessor and successor of a rectangle in the visitation sequence. If R_i is followed by R_j, a finite set of disc handoff points is sampled uniformly in their shared region or along their opposing boundaries. An entry to the first rectangle must also be reachable from q_0, and an exit from the final rectangle is selected among its remaining feasible boundary candidates. Each candidate specifies the disc center, base heading, and reference arm angle rather than only a boundary point.

For disc points on the left, right, lower, and upper boundaries, the nominal headings are π, 0, $3\pi/2$, and $\pi/2$, respectively, so that the base in its retracted configuration lies outside the corresponding rectangle boundary. This rule generates nominal candidates only. Near dense obstacles or narrow boundaries, the externally positioned base may still enter unavailable space and therefore cannot be accepted without checking.

Given a boundary pose $(\mathbf{p}_d, \theta)$ for the disc center, the base center in the retracted reference configuration is recovered as

$$\mathbf{p}_c = \mathbf{p}_d + (d_p + L)\mathbf{e}(\theta). \tag{10}$$

The state and its connection from the preceding primitive are then checked against Eqs. (3a) and (3b). If a check fails, the handoff point is moved along the same boundary, followed by trials on an adjacent boundary and with the alternative heading. Only candidates with a feasible complete connection enter the local cost comparison. If a visit admits no feasible entry–exit pair, the corresponding global sequence cannot be instantiated.

3.5.2 Starting Corner, Strip Width, and Swing Amplitude

The swing amplitude is the half-amplitude of the arm angle about its neutral position while one strip is covered. It is denoted ϕ_w and is determined by the strip width, the disc radius, and the arm length. A work strip is one working band covered by the disc as it travels back and forth along one side of the rectangle. Let ℓ_R and h_R be the rectangle dimensions parallel and perpendicular to the work strips. ℓ_R is the travel length of one work strip, and h_R is the transverse width crossed together by the strips. The local planner uses the longer axis by default to reduce lane changes, but also generates a solution along the other axis. Across the two sweep directions and their feasible starting corners, it compares the total time proxy for entry connection, coverage, lane changes, and exit connection. Thus h_R always denotes the transverse extent crossed collectively by the strips, whereas ℓ_R is the travel length of one strip.

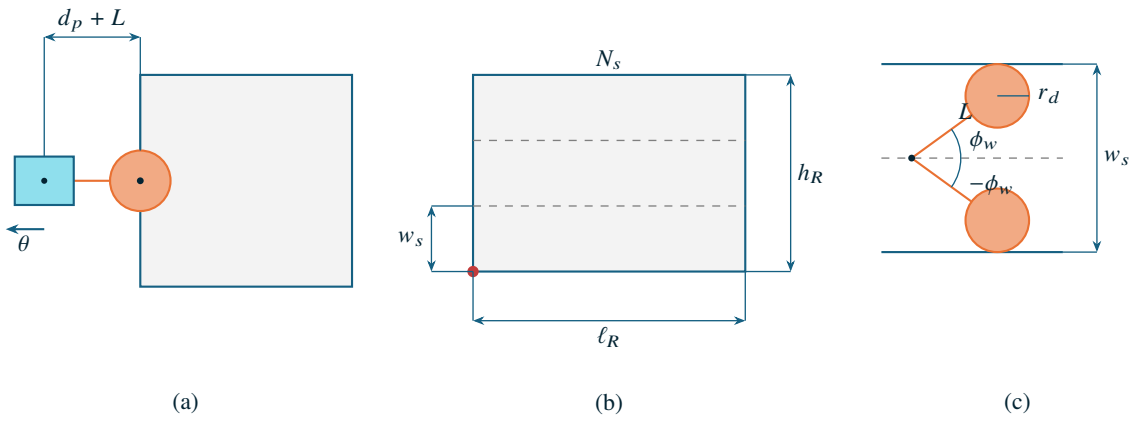


Figure 8: Entry pose, work strips, and swing amplitude ϕ_w for one rectangle.

For each sweep direction, the planner filters the four corners to retain starting points reachable from the entry and connectable to the exit; the parity of the strip count determines the coverage endpoint. It selects the feasible combination with the minimum sum of non-working motion and local coverage time. Entry, sweep direction, starting corner, strip-count parity, and exit therefore jointly determine the primitive; the start is not chosen merely as the nearest corner in Euclidean distance.

Coverage inside the rectangle follows a Boustrophedon pattern, but the work width is not fixed. It is determined by the maximum usable arm angle and disc radius. For arm limits $[\phi_{\min}, \phi_{\max}]$, the symmetric usable angle is

$$\phi_s = \min\{|\phi_{\min}|, |\phi_{\max}|, \pi/2\}. \tag{11}$$

Over one full left–right swing, the lateral width approximately reachable by the disc-center path, including disc radius, is

$$w_{\max} = 2(L \sin \phi_s + r_d). \tag{12}$$

For transverse rectangle dimension h_R, the required number of strips and actual strip width are

$$N_s = \left\lceil \frac{h_R}{w_{\max}} \right\rceil, \qquad w_s = \frac{h_R}{N_s}. \tag{13}$$

The half-amplitude required to cover a strip of width w_s is

$$\phi_w = \begin{cases} 0, & w_s \le 2r_d, \\ \arcsin\left(\dfrac{w_s/2 - r_d}{L}\right), & w_s > 2r_d, \end{cases} \tag{14}$$

clipped to $[0, \phi_s]$. This prevents systematic gaps between adjacent strips due to insufficient swing amplitude.

3.5.3 Coupling Between Base and Arm Speeds

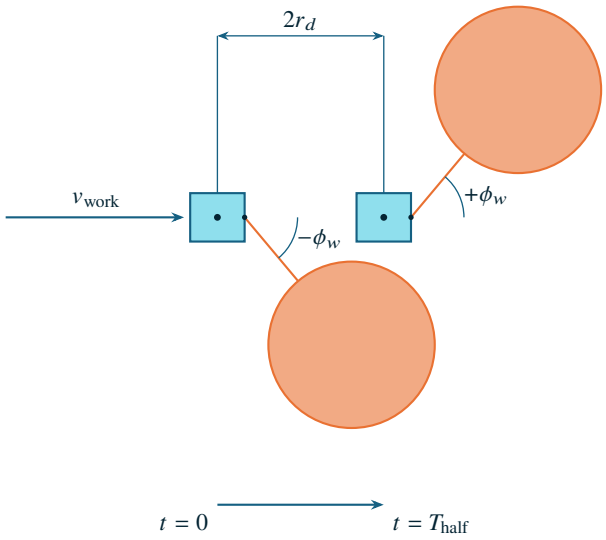


Figure 9: During one half-swing the base advances one disc diameter, so adjacent sweeps meet.

Each strip combines straight base motion with periodic arm swinging. Let the maximum arm angular velocity be $\omega_{a,\max}$ and maximum base linear velocity $v_{\max}$. The time for the arm to move from $-\phi_w$ to $+\phi_w$ is

$$T_{\text{half}} = \frac{2\phi_w}{\omega_{a,\max}}. \tag{15}$$

The base work speed cannot be chosen from the mechanical limit alone if consecutive half-sweeps are to overlap sufficiently in the direction of travel. The method uses the following heuristic upper bound. If $\phi_w = 0$, let $v_{\text{safe}} = +\infty$; otherwise,

$$v_{\text{safe}} = \frac{2r_d}{T_{\text{half}}}. \tag{16}$$

The work speed is

$$v_{\text{work}} = \min(v_{\max}, v_{\text{safe}}). \tag{17}$$

When the arm is slow or the amplitude large, Eq. (17) reduces the base speed so that the forward distance between adjacent half-sweeps does not exceed the disc diameter. This is a conservative rule derived from local sweep spacing, not a proof of complete coverage for every continuous arm trajectory. The actual swept area is therefore still checked after local motion generation by the procedure in Section 3.7. The same base–arm velocity coupling underlies the later experiments.

3.5.4 Time Model for First and Repeated Visits

The first-visit time in Eq. (18) has four terms: strip travel, lane changes, in-place turns, and arm adjustments. Handoff time between rectangles lies in the edge cost c_{ij} and does not enter T_{cover}; travel toward the exit of the present rectangle, when it occurs, is counted in the lane-change or turn term. If strip k has length ℓ_k, lane-change segment k has length d_k, base rotation m is $\Delta\theta_m$, and arm adjustment m is $\Delta\phi_m$, the total coverage time is

$$\begin{aligned} T_{\text{cover}}(R; q^{\text{in}}, q^{\text{out}}) = &\sum_{k=1}^{N_s} \frac{\ell_k}{v_{\text{work}}} + \sum_{k} \frac{d_k}{v_{\max}} \\ &+ \sum_{m} \frac{|\Delta\theta_m|}{\omega_{\max}} + \sum_{m} \frac{|\Delta\phi_m|}{\omega_{a,\max}}. \end{aligned} \tag{18}$$

The model assumes piecewise-constant velocity and serially accumulates translation, in-place rotation, and arm adjustment. It neglects acceleration and deceleration and allows in-place turns wherever the collision constraints permit. Equation (18) is therefore a kinematic proxy for candidate ranking and nominal duration, not a time-optimal solution with actuator dynamics. Entry, exit, and starting corner affect the distances and angles and are consequently explicit arguments.

On the second or later appearance of a rectangle in the sequence, coverage has already been completed. The algorithm generates only an entry-to-exit transfer with time

$$\begin{aligned} T_{\text{transit}}(R; q^{\text{in}}, q^{\text{out}}) = &\sum_{k} \frac{d_k}{v_{\max}} + \sum_{m} \frac{|\Delta\theta_m|}{\omega_{\max}} \\ &+ \sum_{m} \frac{|\Delta\phi_m|}{\omega_{a,\max}}. \end{aligned} \tag{19}$$

A repeated visit contributes no new work coverage but has nonzero execution time. Every straight segment, in-place turn, and arm adjustment must also pass the adaptive collision check in Eq. (3a); failed entry–exit pairs are excluded from subsequent search. Allowing transit through covered rectangles makes a connected graph continuously traversable, while Eq. (19) penalizes excessive revisits in the global route cost.

3.6 Global Route Planning and Full-Trajectory Assembly

The connected graph G gives feasible transfers between rectangles but does not specify an order from the initial robot state. Let

$$\Pi = (R_{\pi_1}, R_{\pi_2}, \ldots, R_{\pi_K}) \tag{20}$$

be a visitation sequence. It must satisfy three conditions: every $R \in \mathcal{R}$ appears for a first visit at least once; every consecutive pair corresponds to an edge of G; and at least one entry pose of the first rectangle is reachable from q_0 through a collision-free initial connection. A rectangle may appear more than once so the robot can transit through a covered region to reach an unvisited branch. Before sequence generation, the algorithm attempts to connect q_0 to entry candidates of each rectangle and stores the minimum feasible initial cost as $T_{\text{start}}(q_0, R_i)$. A rectangle without such a connection cannot start the sequence.

For visit k, let q_k^{in} and q_k^{out} be the selected entry and exit states, and set $I_k = 1$ when R_{π_k} appears for the first time, otherwise $I_k = 0$. The sequence cost is

$$\begin{aligned} J(\Pi) = T_{\text{start}}(q_0, R_{\pi_1}) + \sum_{k=1}^{K} \big[I_k T_{\text{cover}}(R_{\pi_k}; q_k^{\text{in}}, q_k^{\text{out}}) \\ + (1 - I_k) T_{\text{transit}}(R_{\pi_k}; q_k^{\text{in}}, q_k^{\text{out}}) \big] + \sum_{k=1}^{K-1} c_{\pi_k \pi_{k+1}}. \end{aligned} \tag{21}$$

The handoff states of adjacent edges must be consistent. Coverage cost is incurred only on a rectangle's first visit; repeated visits incur transit cost only. Equation (21) is a time proxy for comparing candidate routes, not an exact traveling-salesman objective.

Two complementary graph heuristics generate candidate sequences. The first is a multi-start randomized nearest-neighbor search. Beginning at a rectangle with a feasible initial connection, each step gathers unvisited neighbors, orders them by a weighted sum of c_{ij} and their remaining unvisited degree, and randomly chooses from the first k_{rcl} elements of this restricted candidate list. A trial is discarded if it reaches a node with no unvisited neighbor before covering all rectangles. The number of trials, k_{rcl}, and the random seed belong to Ψ_P; fixing them reproduces the experimental candidate set.

The second heuristic is an open depth-first walk on a weighted minimum spanning tree. Edge weights are c_{ij}. For each feasible root, the tree distance from the root to every node is computed, and the root-to-farthest-node path is designated as the final non-backtracking branch. All other side branches are visited in ascending edge-weight order with a return to the parent; the walk then follows the final branch to its terminal node and stops. Parent nodes and covered rectangles may therefore reappear. This provides a deterministic feasible candidate on a connected graph, but is only a feasibility fallback and is not claimed to be globally optimal in the original graph.

All candidates are first compared by Eq. (21) and then instantiated using the local primitives. Beginning with the initial connection from q_0, the algorithm creates a coverage primitive for every first visit, a transit primitive for every repeat, and a handoff motion between adjacent rectangles. A candidate is retained only if all endpoints agree and every segment passes Eqs. (3a) and (3b). The valid candidate with the smallest J is selected; its segments are shifted in time and concatenated to produce

$$q : [0, T_{\text{exec}}] \to \mathcal{Q}_F. \tag{22}$$

If no candidate can be instantiated as a complete trajectory from q_0, route planning fails. Algorithm 2 summarizes the procedure.

3.7 Work Coverage and Full-Trajectory Coverage

Coverage is evaluated directly from the continuous trajectory rather than from a video frame rate. The trajectory is partitioned into timed segments for local coverage, turns, lane changes, arm adjustment, transit, and inter-rectangle handoff. A segment is marked as effective work only while the disc is in its working state, the base moves along a rectangular work strip, and the arm executes the prescribed swing. Entry connections, lane changes, in-place rotations, arm deployment or retraction, repeated transit visits, and inter-rectangle handoffs are marked as non-working.

For any segment s, let the disc-center path be

$$\mathbf{p}_d^s(\tau), \qquad \tau \in [0, T_s]. \tag{23}$$

The disc-swept region of the segment is

$$C_s = \bigcup_{\tau \in [0, T_s]} \left\{ \mathbf{z} : \|\mathbf{z} - \mathbf{p}_d^s(\tau)\|_2 \le r_d \right\}. \tag{24}$$

Algorithm 2 Global route planning and trajectory assembly

Require: Rectangles $\mathcal{R}$, connected feasible graph G, initial state q_0, local primitives $\mathcal{L}$, parameters Ψ_P
Ensure: Visit sequence Π and continuous trajectory $q(t)$, or route failure

$S \leftarrow \text{FeasibleStarts}(q_0, \mathcal{R}, G)$
if $S = \varnothing$ **then**
    return route failure
end if
$C \leftarrow \varnothing$
for each feasible start $s \in S$ **do**
    $\mathcal{P} \leftarrow \text{RandomizedNearestNeighbor}(G, s, \Psi_P)$
    $\mathcal{P} \leftarrow \mathcal{P} \cup \{\text{OpenMstDfs}(G, s, c)\}$ ▷ farthest branch last
    for each visitation sequence $\Pi \in \mathcal{P}$ **do**
        if Π does not visit every rectangle in $\mathcal{R}$ **then**
            continue
        end if
        $a \leftarrow \text{CompatibleEntryExitStates}(\Pi, G)$
        if $a \neq$ failure **then**
            $C \leftarrow C \cup \{(\Pi, a, J(\Pi, a))\}$ ▷ Eq. (21)
        end if
    end for
end for
for each (Π, a, J) in C, in increasing J **do**
    $q \leftarrow \text{AssembleTrajectory}(q_0, \Pi, a, \mathcal{L})$
    Add initial and inter-rectangle connectors; cover first visits and transit repeats.
    if q is continuous and satisfies Eqs. (3a) and (3b) **then**
        return (Π, q)
    end if
end for
return route failure

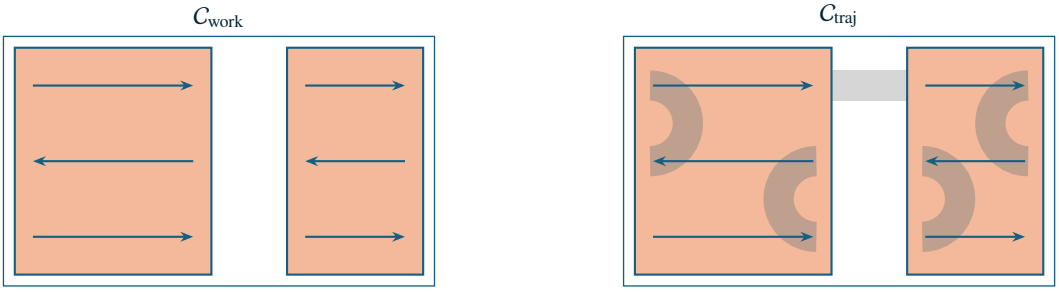


Figure 10: Work coverage counts disc sweeps on work strips; full-trajectory coverage also counts sweeps during transfer and adjustment.

Let $\mathcal{S}_{\text{work}}$ be the set of effective-work segments and $\mathcal{S}_{\text{all}}$ the set of all motion segments. The method distinguishes

$$C_{\text{work}} = \bigcup_{s \in \mathcal{S}_{\text{work}}} C_s, \qquad C_{\text{traj}} = \bigcup_{s \in \mathcal{S}_{\text{all}}} C_s. \tag{25}$$

The former contains only work-labeled disc sweeps, whereas the latter also includes disc motion during transfer, lane changes, and pose adjustment. Their respective coverage ratios are

$$\begin{aligned} \rho_{\text{work}} &= \frac{|\Pi_p(C_{\text{work}}) \cap \mathcal{P}_F|}{|\mathcal{P}_F|}, \\ \rho_{\text{traj}} &= \frac{|\Pi_p(C_{\text{traj}}) \cap \mathcal{P}_F|}{|\mathcal{P}_F|}, \end{aligned} \tag{26}$$

where $\Pi_p(\cdot)$ rasterizes a continuous world-coordinate region into a pixel set. The primary coverage measure is ρ_{work} because it corresponds to explicitly labeled effective-work motion. The value ρ_{traj} describes how much free area is swept over the complete motion. If the disc remains in contact with the floor during non-working transfers, it captures additional physical sweeping; if the disc is lifted, it is only an auxiliary measure of path reach or potential sweep and must not be conflated with effective work coverage.

To avoid dependence on an arbitrary sampling frame rate, $\mathbf{p}_d^s(\tau)$ is recursively subdivided. A time interval $[\tau_a, \tau_b]$ is accepted only when the endpoint chord length is no greater than η_ℓ and the deviation of the midpoint from that chord is no greater than η_c. Both are chosen no larger than $s_p/2$, bounding discretization error by map resolution. Let the final subdivision points be $\mathbf{c}_0, \ldots, \mathbf{c}_m$. Rasterizing the radius-r_d capsule between every adjacent pair yields

$$\widehat{C}_s = \Pi_p\left(\bigcup_{j=0}^{m-1} \text{Capsule}(\mathbf{c}_j, \mathbf{c}_{j+1}, r_d)\right). \tag{27}$$

The capsule union is constructed in world coordinates before conversion by Π_p, avoiding a type mismatch between continuous regions and pixel sets. This method is more stable than drawing discs only at fixed-rate samples, especially during fast arm motion.

Finally, nominal execution time and planning time are

$$T_{\text{exec}} = \sum_{s \in \mathcal{S}_{\text{all}}} T_s, \qquad T_{\text{plan}} = t_{\text{eval,end}} - t_{\text{raster,start}}. \tag{28}$$

The first follows the piecewise-constant and serial-timing assumptions used for the local primitives. The second is measured as wall-clock time in one process, from conservative-grid generation through continuous-sweep evaluation. It includes decomposition, connectivity completion, route search, trajectory assembly, and coverage computation, but excludes map-file reading, output writing, and visualization. Thus ρ_{work}, ρ_{traj}, T_{exec}, and T_{plan} match the output definitions in Fig. 4.

4 Experiments and Results

Three groups of experiments are designed to evaluate the proposed swing-arm CPP method. The first compares BiRect with Fields2Cover to verify that, on obstacle maps, active swing-arm motion gives executable coverage and a higher full-trajectory coverage ratio than fixed-width strips. The second compares rectangular-candidate scan strategies to determine how bidirectional candidates affect coverage quality and redundancy in complex spaces. The third varies base speed at a fixed arm angular velocity to expose how base–arm velocity coupling affects coverage and execution time.

All three groups use two base maps. `simple` contains four regular rectangular obstacles and tests basic obstacle avoidance and coverage. `complex` contains densely distributed small obstacles and better resembles a construction floor partitioned by columns, openings, and local structures. The scan-strategy experiment additionally uses three directional maps:

- `scan_comb_corridors`: horizontal comb-like corridors;
- `scan_mixed_weave`: interwoven horizontal and vertical structures; and
- `scan_offset_rooms`: rooms connected with offsets.

These structures distinguish the effects of row-only, column-only, and bidirectional scanning.

Every map is a binary occupancy image, with free and unavailable pixels as defined above. The resolution is 0.5 m/pixel. Except in the speed experiment, the same parameters are used throughout: disc radius $r_d = 0.75$ m, arm length $L = 1.20$ m, base width 0.80 m, maximum base linear velocity 2.00 m/s, maximum base angular velocity 30°/s, and maximum arm angular velocity 90°/s. Outside the speed experiment these three limits are used together. The actual work speed is still $v_{\text{work}} = \min(v_{\max}, v_{\text{safe}})$ from Section 3.5.3, so the limits are not the travel speed of every strip. In the speed experiment the arm angular velocity stays 90°/s and the base angular velocity stays 30°/s, while the base linear velocity takes the six values 0.5, 1.0, 1.5, 2.0, 2.5, and 3.0 m/s. The minimum rectangle side length is 2.5 m, the greedy-selection weights are $\alpha = 0.5$ and $\beta = 2.0$, and the random seed is 42.

4.1 Executability and Coverage against Fields2Cover

Fields2Cover is the fixed-width representative from Section 2, and it is used here to verify executable coverage and a higher full-trajectory coverage ratio.

For Fields2Cover, the working width is 1.50 m, equal to the diameter of the proposed method's 0.75 m-radius disc. With base width 0.80 m and safety margin 0.05 m, the original free region is eroded by 0.45 m. Both methods are evaluated over their complete trajectories. For the proposed method, coverage is the area swept by the disc along the base–arm trajectory; for Fields2Cover, it is the fixed-width tool sweep along work, turning, and transfer segments. To align evaluation with the actual planning domain, the Fields2Cover denominator is the eroded free-space area. Its fixed-width sweep uses flat strip ends and mitered path corners. The work-strip speed is 2.0 m/s, while turns and connecting paths use 0.5 m/s.

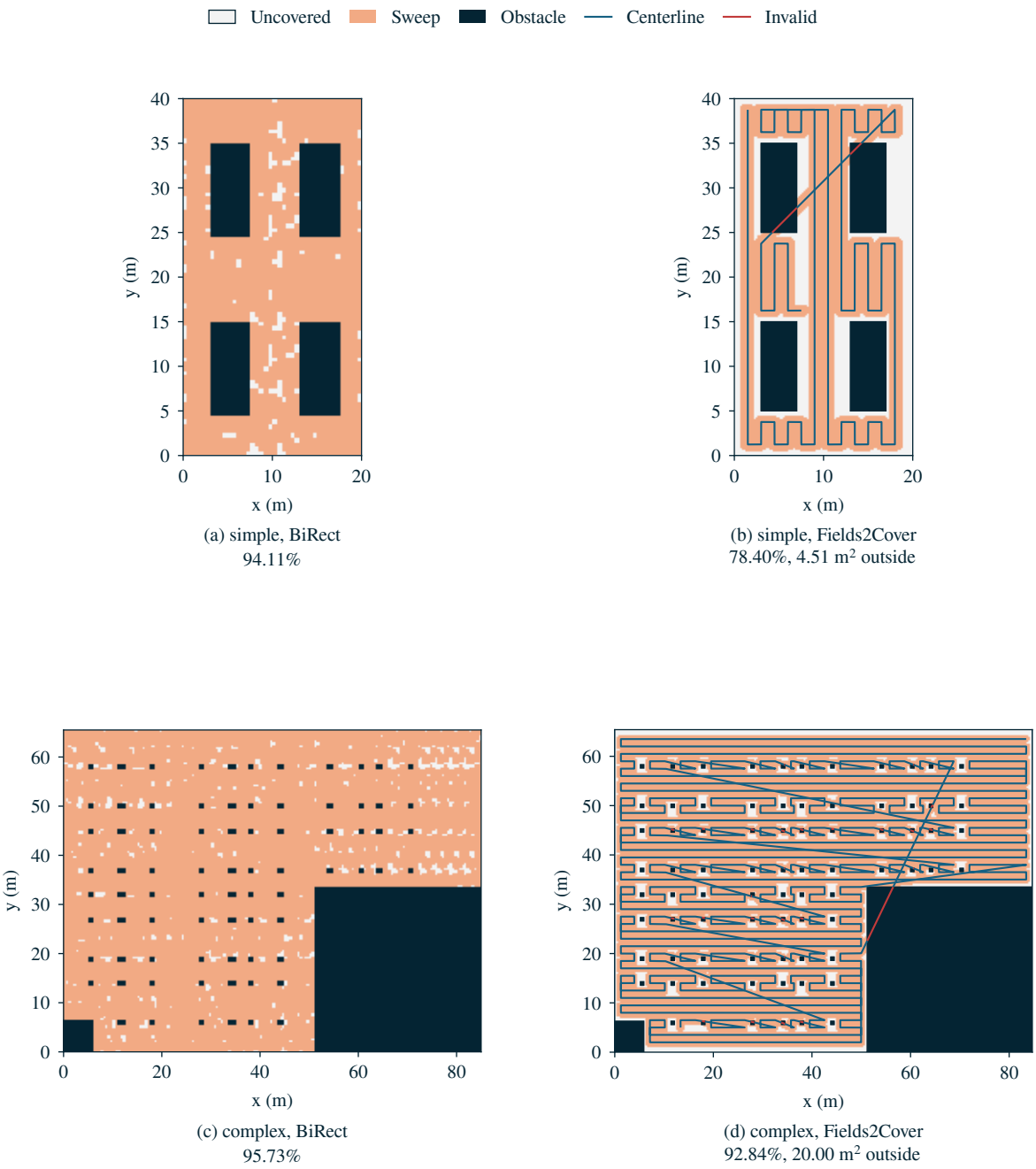


Figure 11: Coverage paths and swept regions for the proposed method and Fields2Cover: panels (a) and (b) show `simple`, and panels (c) and (d) show `complex`.

Table 2: Comparison between the proposed method and Fields2Cover.

Map	Method	Evaluation area (m^2)	Covered area (m^2)	Coverage ratio (%)	Execution/planning time (s/s)	Path check
`simple`	**BiRect**	611.00	575.00	**94.11**	854.73/1.298	**Passed**
	Fields2Cover	533.17	418.00	78.40	295.50/0.075	4.51 m^2 outside
`complex`	**BiRect**	4294.00	4110.50	**95.73**	5844.73/17.981	**Passed**
	Fields2Cover	4046.79	3756.85	92.84	2883.81/6.339	20.00 m^2 outside

The evaluated areas in Table 2 differ between methods, so their coverage ratios should not be interpreted as same-domain percentage-point differences. The Fields2Cover routes fail the path check; their execution

and planning times are reported only for the generated outputs.

On `simple`, Fields2Cover has shorter generation and nominal execution times with a 1.50 m fixed width equal to the disc diameter. Its complete trajectory sweeps 418.00 m^2 (78.40%), compared with 575.00 m^2 (94.11%) for the proposed method. Its connecting segments, however, sweep 4.51 m^2 outside the admissible region. The recorded 295.50 s execution time and 0.075 s planning time therefore describe an invalid generated path rather than an executable solution. This illustrates the principal benefit of the swing-arm configuration: the base need not drive along every strip that the disc must cover. Arm motion extends the covered region and creates a wider time-varying sweep within the same free space.

On `complex`, the proposed method covers 4110.50 m^2, or 95.73% of its evaluation domain. Fields2Cover sweeps 3756.85 m^2 (92.84%) along the complete trajectory. Yet its connecting path for the 1.50 m tool leaves the free space, creating 20.00 m^2 of invalid base sweep; the corresponding 2883.81 s execution time and 6.339 s planning time are likewise not valid baseline scores. The problem is concentrated near dense local obstacles. Although individual swaths fit inside the eroded region, curvature-constrained connections between swaths require additional room. Without a broad transfer area analogous to an agricultural headland, these connections can cross the boundary of the original free space. Active arm sweeping partially decouples base travel from end-disc coverage and preserves greater coverage capability and path continuity in such complex regions.

4.2 Effect of the Rectangle-Candidate Scan Strategy

The preceding comparison shows that an active arm can create coverage patterns unlike those of fixed-width strips. Realizing this advantage consistently also requires a decomposition that represents free space appropriately. A candidate generator biased toward one scan direction can produce fragmented or heavily overlapping rectangles; even when the resulting trajectory is executable, unnecessary transfers may follow. This experiment traces how scan direction propagates through the rectangle set, visitation sequence, and complete trajectory.

The compared strategies are row-only, column-only, and row-then-column scanning. Robot parameters, minimum rectangle size, greedy weights, random seed, connectivity completion, visitation ordering, and local coverage generation are fixed. Measures include greedy coverage, final decomposition coverage, decomposition overlap, full-trajectory coverage, and total execution time, distinguishing representation quality from execution performance. Candidates are first generated and greedily selected from a binary map. After hole filling and connectivity completion, the same trajectory generation and disc-sweep evaluation method is applied in every case.

Table 3: Results of three candidate-scan strategies on five maps.

Map	Strategy	Greedy coverage	Final partition coverage	Partition overlap	Trajectory coverage	Total time (s)
`simple`	**row-only**	72.01%	100.00%	34.21%	93.04%	**696.89**
	col-only	72.01%	100.00%	34.21%	93.04%	**696.89**
	row-then-col	**100.00%**	100.00%	**0.00%**	**94.11%**	854.73
`complex`	**row-only**	87.32%	100.00%	75.34%	95.73%	**5844.73**
	col-only	87.32%	100.00%	75.34%	**95.90%**	5896.03
	row-then-col	87.32%	100.00%	75.34%	95.73%	**5844.73**
Comb corridors	row-only	87.18%	100.00%	34.46%	93.82%	2065.09
	col-only	86.53%	100.00%	31.46%	93.94%	2032.55
	row-then-col	**96.22%**	100.00%	**27.13%**	**94.01%**	**1986.42**
Mixed weave	row-only	84.23%	100.00%	47.89%	95.26%	1532.65
	col-only	83.17%	100.00%	52.59%	95.49%	1575.63

Map	Strategy	Greedy coverage	Final partition coverage	Partition overlap	Trajectory coverage	Total time (s)
	row-then-col	**96.76%**	100.00%	**40.74%**	**95.70%**	**1522.77**
Offset rooms	row-only	82.95%	100.00%	38.78%	91.97%	1496.73
	col-only	81.81%	100.00%	36.74%	91.51%	1443.31
	row-then-col	**99.14%**	100.00%	**24.46%**	**92.06%**	**1374.38**
Five-map mean	row-only	–	–	–	93.96%	2327.22
	col-only	–	–	–	93.98%	2328.88
	row-then-col	–	–	–	**94.32%**	**2316.60**

The two general maps first establish basic behavior in non-directional scenes. On `simple`, row-then-column achieves the highest full-trajectory coverage (94.11%) and reduces partition overlap to zero, but its 854.73 s execution time exceeds the 696.89 s of either one-direction strategy. In this regular obstacle scene, bidirectional scanning trades longer execution for higher coverage. On `complex`, row-then-column and row-only select the same rectangles and give identical results. Column-only provides slightly higher coverage (95.90%), whereas row-only and row-then-column tie for the shortest total time (5844.73 s). Bidirectional scanning therefore does not dominate every metric in every scene. When free space has a pronounced principal direction or row candidates already capture the structure, a one-direction strategy may be faster or marginally improve coverage.

The three directional maps then isolate scan-direction bias; they are not intended to prove that bidirectional scanning is optimal on arbitrary maps. Horizontal corridors, orthogonal interweaving, and offset connections amplify residual regions created by a single scan direction. On all three maps, row-then-column gives the highest greedy coverage and lowest partition overlap. Column candidates supplement regions that row candidates represent poorly rather than merely increasing the rectangle count. This improvement propagates to execution: row-then-column gives both the highest full-trajectory coverage and shortest time on all three maps, respectively 94.01%/1986.42 s, 95.70%/1522.77 s, and 92.06%/1374.38 s. Their mean is 93.93%/1627.86 s. When a map truly contains residual structures in both directions, complementary row and column candidates reduce directional bias and redundant visitation while increasing useful sweep along the complete trajectory.

Across the five maps, the arithmetic mean of full-trajectory coverage is 94.32% for row-then-column, 93.96% for row-only, and 93.98% for column-only. Its mean execution time of 2316.60 s is also slightly below 2327.22 s and 2328.88 s. It is therefore best characterized as a robust default: it is more likely to preserve high coverage without a substantial time penalty when horizontal and vertical residual structures coexist. For a regular region with a single principal direction, a one-direction strategy may still be preferable under the specific coverage–schedule trade-off. Figure 12 visualizes the directional complementarity on `scan_mixed_weave`.

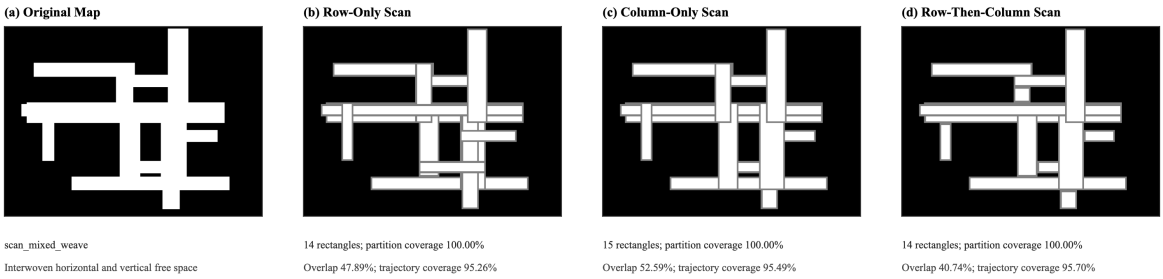


Figure 12: Candidate-scan strategies on `scan_mixed_weave`: (a) original map; (b) row-only decomposition; (c) column-only decomposition; and (d) row-then-column decomposition.

4.3 Effect of Base–Arm Velocity Coupling on Coverage

The first two experiments address route and decomposition quality. Swing-arm coverage also has an execution issue that is less prominent in fixed-width CPP: the same geometric route produces different disc-sweep densities at different base speeds. As the base advances faster, it travels farther between adjacent arm sweeps; slower travel improves coverage but substantially increases task time. Reporting a geometric route without analyzing base–arm velocity coupling therefore cannot establish the coverage quality and time cost of actual execution.

This experiment tests the analysis in Section 3.5.3. At fixed arm angular velocity, increasing base speed should shorten execution but enlarge the spacing between adjacent sweeps and reduce coverage. Keeping the rectangle set, visitation order, and local-work structure fixed isolates the effect of base speed.

The arm angular velocity is fixed at $\omega_a = 90°/\text{s}$. The same planned solutions are executed on `simple` and `complex` with base speed increasing from 0.5 to 3.0 m/s. The ratio of this base–arm velocity coupling is

$$\rho = \frac{v}{\omega_a},$$

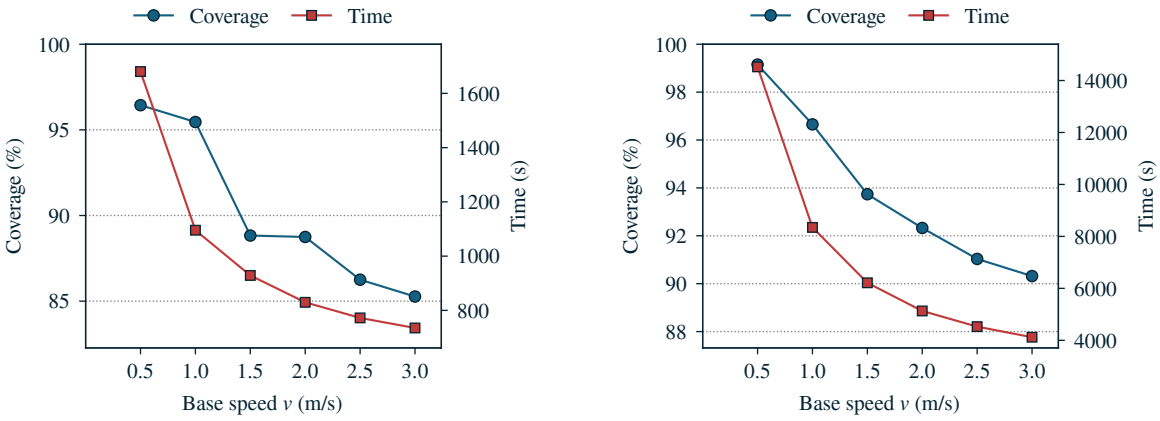
with units of m/degree. It is the base distance traveled while the arm rotates through $1°$. A larger ρ increases forward spacing between adjacent sweeps and makes coverage gaps more likely. Coverage, total time, and time per unit covered area are recorded to expose the quality–efficiency trade-off.

Table 4: Full-trajectory coverage and execution time at different base speeds.

Map	Base speed (m/s)	ρ (m/degree)	Full-path coverage	Total time (s)	Time per covered area (s/m^2)
`simple`	0.5	0.00556	96.44%	1680.45	2.852
	1.0	0.01111	95.46%	1095.53	1.878
	1.5	0.01667	88.83%	928.81	1.711
	2.0	0.02222	88.75%	829.74	1.530
	2.5	0.02778	86.25%	772.26	1.465
	3.0	0.03333	85.27%	735.12	1.411
`complex`	0.5	0.00556	99.15%	14525.14	3.412
	1.0	0.01111	96.65%	8346.21	2.011
	1.5	0.01667	93.74%	6214.22	1.544
	2.0	0.02222	92.33%	5130.64	1.294
	2.5	0.02778	91.03%	4526.86	1.158
	3.0	0.03333	90.32%	4120.11	1.062

Table 4 shows a clear coverage–time trade-off, although the full-trajectory results make it more moderate than an analysis of work strips alone might suggest. Total time decreases monotonically with base speed: from 1680.45 to 735.12 s (56.25%) on `simple` and from 14525.14 to 4120.11 s (71.63%) on `complex`. As ρ increases from 0.00556 to 0.03333 m/degree, coverage decreases from 96.44% to 85.27% and from 99.15% to 90.32%, reductions of 11.17 and 8.83 percentage points. Larger sweep spacing therefore weakens coverage continuity, but no tested condition makes the full-trajectory coverage unusably low.

Time per unit covered area further clarifies the trade-off. It decreases continuously with base speed on both maps: from 2.852 to 1.411 s/m^2 on `simple` and from 3.412 to 1.062 s/m^2 on `complex`, improvements of 50.53% and 68.88%. A speed of 3.0 m/s gives both the shortest total time and lowest time per covered area, at the cost of coverage ratios of 85.27% and 90.32%. Conversely, 0.5 m/s gives the highest coverage on both maps. If both maps must retain at least 95% full-trajectory coverage, 1.0 m/s is the highest tested admissible speed, while already reducing total time by 34.81% and 42.54% relative to 0.5 m/s. Base speed should therefore be selected jointly with arm angular velocity according to a minimum-coverage constraint and schedule objective, rather than treated as an incidental parameter after geometric planning. Smaller ρ favors coverage quality; larger ρ favors time efficiency when a coverage loss of roughly 9–11 percentage points is acceptable.



(a) `simple` (b) `complex`

Figure 13: Effect of base–arm velocity coupling on full-trajectory coverage and execution time.

4.4 Discussion

Relative to fixed-width strips, the swing arm gives executable coverage on obstacle maps. Bidirectional candidates reduce the regions left by a single scan direction. Base speed relative to arm angular velocity determines coverage and execution time.

These findings do not imply unconditional superiority over a fixed-width platform. In regular open areas, a strip-based method can obtain acceptable coverage with a shorter path and lower time. The value of the swing arm is greatest at boundaries a fixed base cannot closely follow, in obstacle-dense regions, and on wet concrete where fewer base maneuvers are desirable. Platform and parameter selection can therefore depend on site structure: large open regions favor efficient fixed-width sweeps, while complex corners and multi-obstacle regions favor arm-assisted coverage. The two could also operate in separate zones of the same floor.

The method has several limitations. First, rectangle candidates and visitation sequences are generated heuristically. They provide feasible and interpretable routes but no global optimality guarantee; many narrow corridors can still increase the rectangle count and repeated transit. Second, the input is a static two-dimensional map and does not explicitly account for workers, material stacks, localization drift, or changes in concrete condition during construction. Third, the speed analysis focuses on coverage density under base linear and arm angular velocities; it does not yet model full chassis dynamics, disc pressure, surface-quality feedback, or control error. Overall, the experiments identify explicit modeling of the end-tool sweep, area decomposition, and base–arm velocity coupling as the features that distinguish swing-arm troweling CPP from conventional fixed-width planning and define the next steps toward field deployment.

5 Conclusion and Future Work

Obstacles and narrow connections in complex construction maps prevent a swing-arm troweling robot from obtaining a continuous, executable coverage route from either a fixed tool footprint or a static reachability region alone. This paper presented a hierarchical CPP method for known two-dimensional environments with obstacles. Bidirectional rectangular candidates, hole filling, and connectivity completion represent the workable area; a feasible graph organizes global visitation; and first visits and repeated transit visits are instantiated as coordinated base–arm work motions and transfer motions. Planning and evaluation preserve the geometry of the base, arm, and circular disc, and coverage and nominal execution time are calculated from the disc's actual sweep along the continuous trajectory. The method thus connects map

decomposition, region visitation, and end-tool coverage verification in one complete pipeline.

On `simple` and `complex`, the proposed method achieves full-trajectory coverage ratios of 94.11% and 95.73%, respectively, and both routes pass the path check. Although the fixed-width Fields2Cover baseline generates shorter paths, its connecting segments leave the valid region by 4.51 m^2 and 20.00 m^2 on the two maps, so the associated nominal times cannot be treated as executable-route results. The comparison demonstrates the practical value of adjustable relative motion between base travel and disc operation in dense obstacle fields. In the five-map scan-strategy study, row-then-column candidates give the highest mean full-trajectory coverage (94.32%) and the lowest mean execution time (2316.60 s), indicating that complementary row and column directions reduce representation bias; a single-direction strategy can nevertheless be preferable in some regular scenes. The speed study further shows that increasing base speed relative to arm angular velocity enlarges the spacing between sweeps and creates a coverage–time trade-off. Route geometry and execution parameters must therefore be selected together.

The current method relies on heuristic decomposition and visitation ordering and does not guarantee global optimality. It assumes a known static two-dimensional map, and its evaluation does not yet include complete chassis dynamics, localization error, disc pressure, or surface-quality feedback. Future work will jointly optimize decomposition and visitation, introduce online perception and local replanning, and validate trajectory execution and finishing quality on a physical swing-arm troweling robot. The framework may also serve as a decision-support tool for construction-map partitioning and base–arm motion configuration.

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